

Feasibility-Aware Quantum–Classical Benchmarking of a HUBO Formulation for Occlusion-Aware UAV Trajectory Optimization

Higher-Order Binary Optimization · Simulated Annealing · QAOA · A* Baseline

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We introduce a **quantum-compatible HUBO formulation** for occlusion-aware UAV trajectory optimization and a **feasibility-aware benchmarking methodology** to evaluate it rigorously. Classical baselines are deliberate: current quantum hardware limits make strong classical references essential, while the higher-order formulation and reduced-instance QAOA experiment chart a clear path toward future **D-Wave / QAOA hardware studies** with a moving target, where occlusion becomes intrinsically cubic—beyond A*'s shortest-path reach.

1 THE PROBLEM

UAV target tracking in cluttered environments must simultaneously:

- Avoid static and dynamic **obstacles**
- Maintain continuous **line-of-sight** to a target (occlusion-free)
- Minimise path length and dwell time in buffer zones

Masouei et al. solved this as continuous multi-convex MPC. We recast the **same problem class** as a discrete higher-order binary problem for quantum solvers — enabling future D-Wave hardware runs without algorithm redesign.

Key insight: with a moving target, occlusion becomes intrinsically cubic in binary variables — a natural HUBO, no quadratisation possible without severe overhead.

2 HUBO FORMULATION

BINARY ENCODING

$x[t, v] = 1$ if UAV occupies cell v at time t

Seven-term Hamiltonian:

$$H = H_{\text{init}} + H_{\text{target}} + H_{\text{obst}} + H_{\text{col}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{turn}}$$

Occlusion (Bresenham LOS):

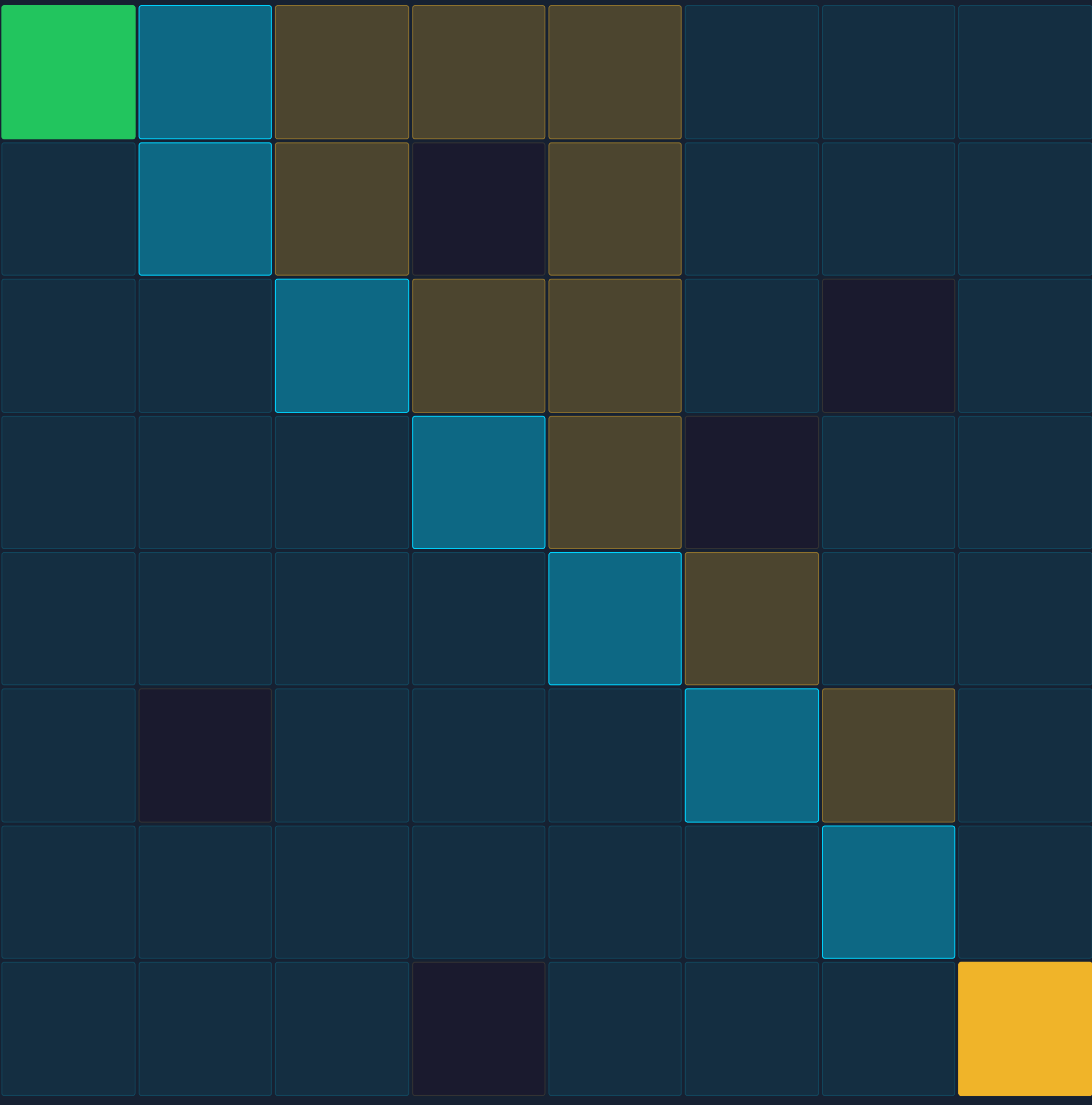
$$H_{\text{occ}} = \sum_{\text{occ}} \sum_{t=0}^{T-1} x[t, u] \cdot \text{LOS}(u, \text{tgt}) \cdot n \cdot c$$

Genuinely cubic proximity (buffer-cell ingesting):

$$H_{\text{prox}} = \sum_{\text{prox}} \sum_{t=0}^{T-3} \sum_b x[t+1, v_b] \cdot x[t, v_a] \cdot x[t+1, v_b]$$

Why HUBO, not QUBO? Quadratising the cubic term requires ~3,172 ancilla qubits (~30% more variables) with dense couplings that destroy graph sparsity. Native HUBO is structurally superior.

OCCLUSION COST SURFACE — 8x8 GRID



3 FEASIBILITY-AWARE BENCHMARKING

Raw HUBO energy **miscalc** — annealers lower energy by violating soft constraints. Four safeguards:

- Energy decomposition**
Hard penalty vs. soft cost tracked separately; never summed into a single scalar.
- Independent constraint check**
Decode binary solution + verify constraints directly — not via energy thresholds.
- Matched sampling budgets**
Equal reads/runs across all solvers for fair wall-clock comparisons.
- Penalty-weight sweep**
Exposes the constraint-exploitation regime; a single λ would misreport feasibility.

4 TWO-TIER EXPERIMENTAL DESIGN

State-vector QAOA scales as 2^n — infeasible at $n=960$. We split into two tiers:

TIER 1 — FULL SCALE

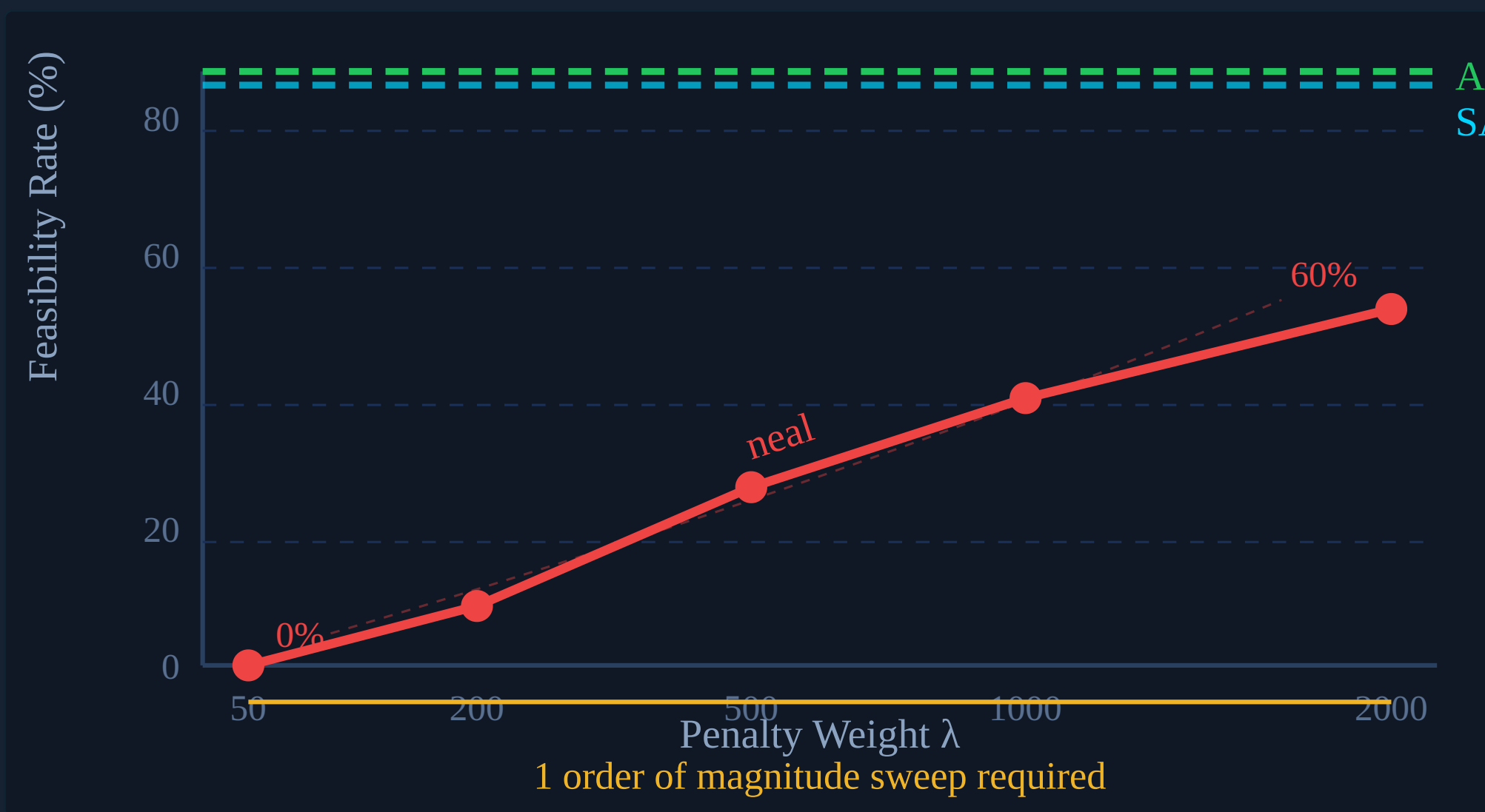
8x8 grid: 960 binary variables
Solvers: A*, SA, **real**
Classical feasibility study

TIER 2 — QUANTUM

2x2 grid: 8 binary variables
Add: Qiskit QAOA
Convergence verification

Classical A* on the same grid with occlusion-weighted cell cost grounds the HUBO against a conventional planner — verifying the formulation captures sensible planning, not a benchmark artifact.

HEADLINE: CONSTRAINT-EXPLOITATION CURVE



real feasibility climbs 0% → 60% as λ 50 → 2000. General-purpose annealers need penalty weights an **order of magnitude above objective scale** to respect one-hot HUBO structure.

5 RESULTS: 8x8 INSTANCE

Metric	A*	SA	NEAL
Soft cost	317	303 ✓	496
Feasibility	100%	100%	22%
Reaches target	Yes	14%	8%
Viability	90%	—	—
Runtime/sample	10 ms	47.8 s	0.26 s

A* and SA reach comparable quality on this static instance. A* is ~4,700x faster. **neal** is fast but only **22% feasible** at the default λ — constraint exploitation confirmed.

6 TIER 2 & QAOA VERIFICATION

On the 2x2 instance (8 variables), **all four solvers** find the same global optimum:

$$E^* = -118.17$$

QAOA serves as a **correctness oracle**, not a speed benchmark. It is **2,750x slower** and does not scale — but confirms the HUBO energy landscape is correctly formulated.

QAOA Circuit

Qiskit · state-vector sim · p=1 layer · m=8 vars · 2ⁿ=256 states

Why This Matters

Validates HUBO structure before D-Wave hardware run

7 KEY TAKEAWAYS

- The **HUBO** value is not raw speed on static instances — it is **quantum compatibility** and structural extensibility to a moving target, where occlusion becomes intrinsically cubic.
- Feasibility-aware methodology** is essential: a single penalty weight's misreporter solver quality, a sweep over 40x range reveals true constraint-exploitation behaviour.
- Native **HUBO** avoids **3,172 ancilla qubits** required by QUBO quadratisation — directly relevant to near-term hardware qubit budgets.
- Classical **A*** and **SA** are strong baselines on static grids; the quantum path gains value as the target moves and the problem becomes genuinely non-polynomial.

8 FUTURE WORK

[MOVING TARGET](#) [REAL D-WAVE HARDWARE](#) [VS. CONTINUOUS MPC](#) [3D GRID EXTENSION](#)

The moving-target extension introduces time-varying LOS sets, making H_{occ} intrinsically degree-3 in the binary encoding — validating native HUBO as the correct representation class for this problem family.